From Fourier Series to Spherical Harmonics

View-Dependent Appearance in 3D Gaussian Splatting (2026/06/03)

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Contents

Section I: Background and Fundamental Theory

From ordinary signals to basis-function expansions

Background

- Fourier analysis originated from the study of heat transfer in the early 19th century.
- Joseph Fourier proposed that complex periodic signals could be represented by sums of simple sine and cosine waves.
- This idea later became a foundation of signal processing, communication systems, image analysis, and modern visual computing.
- The core insight is simple: complicated signals can be decomposed into simpler basis functions.

Complex Signal $\rightarrow$ Sine and Cosine Bases $\rightarrow$ Frequency Coefficients

Fourier Series

A periodic signal can be represented as a weighted sum of sine and cosine functions:

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos(nx) + b_n \sin(nx)\right)$$

where:

- a_0 represents the average level of the signal.
- a_n and b_n are Fourier coefficients.
- $\cos(nx)$ and $\sin(nx)$ are basis functions.
- n controls the frequency of each component.

Approximating a Signal by Increasing n

- Using only low-frequency terms gives a rough and smooth approximation.
- Adding more terms allows the representation to capture sharper changes and finer details.

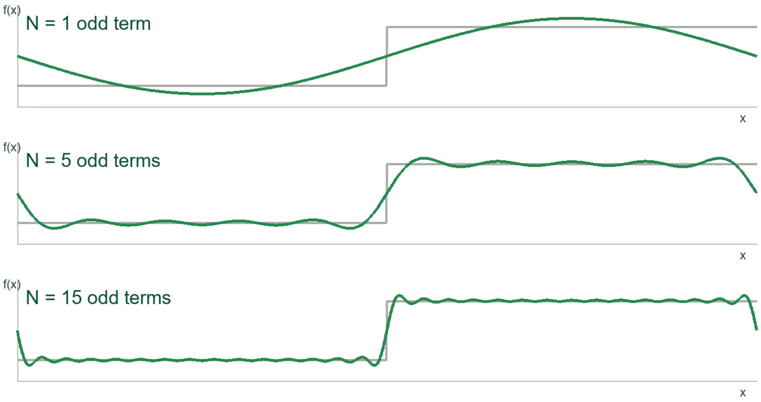

Figure: Square-wave approximation with increasing Fourier terms

Fourier Ideas in 2D Images

- For object boundaries, a closed contour can also be represented in polar form:

$$r = r(\theta)$$

- Then Fourier series can approximate this boundary by adding more frequency components.

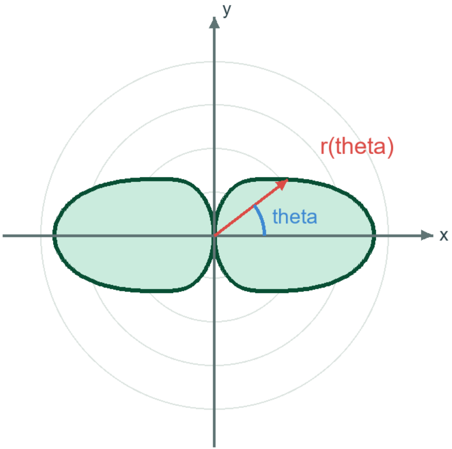

Figure: A dumbbell-shape in polar coordinates

Two-Variable Fourier Series

A two-variable signal can be written as:

$$z = f(x, y)$$

where x and y are spatial coordinates, and z is the signal value or surface height.
For a periodic 2D signal, the Fourier series can be written as:

$$f(x, y) = \sum_{m=-M}^{M} \sum_{n=-N}^{N} c_{mn} e^{j(mx+ny)}$$

Equivalently, it can be expressed using sine and cosine terms:

$$\begin{aligned} f(x, y) \approx \sum_{m=0}^{M} \sum_{n=0}^{N} \big[& A_{mn} \cos(mx)\cos(ny) + B_{mn} \cos(mx)\sin(ny) \\ & + C_{mn} \sin(mx)\cos(ny) + D_{mn} \sin(mx)\sin(ny) \big] \end{aligned}$$

Two-Variable Fourier Series for Surface Signals

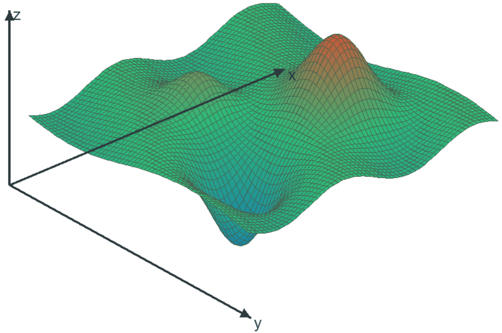

Figure: Two-variable surface signal

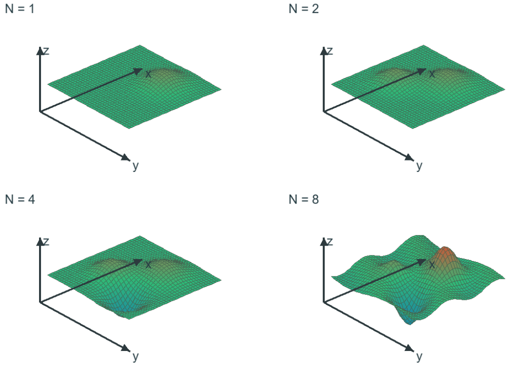

Figure: Increasing 2D Fourier terms

Section II: Spherical Harmonics

Extending Fourier representation from a plane to a sphere

Fourier-Like Expansion in Spherical Coordinates

In spherical coordinates, a 3D surface can be described by a radius function:

$$r = r(\theta, \phi)$$

where:

- r is the distance from the origin.
- θ is the polar angle.
- ϕ is the azimuth angle.

The corresponding Cartesian coordinates are:

$$x = r(\theta, \phi) \sin\theta \cos\phi$$

$$y = r(\theta, \phi) \sin\theta \sin\phi$$

$$z = r(\theta, \phi) \cos\theta$$

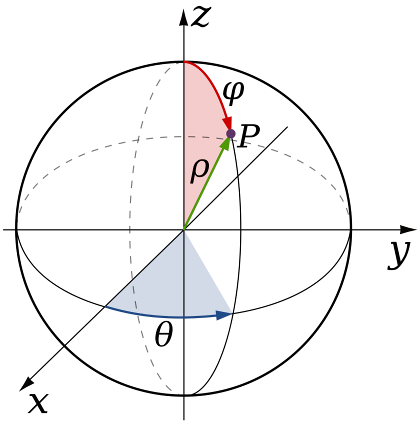

Figure: Spherical coordinates with radius, polar angle, and azimuth angle

Fourier-Like Expansion in Spherical Coordinates

Therefore, the basis expansion changes its domain from a plane to a sphere.
For a 2D signal on a plane:

$$f(x,y) \approx \sum_{m=-M}^{M} \sum_{n=-N}^{N} c_{mn} \Phi_{mn}(x,y)$$

For a directional signal on a sphere:

$$r(\theta,\phi) = \sum_{l=0}^{L} \sum_{m=-l}^{l} c_l^m Y_l^m(\theta,\phi)$$

where:

- l is the degree, controlling the frequency level.
- m is the order, controlling angular variation.
- $Y_l^m(\theta,\phi)$ is a spherical harmonic basis function.
- c_l^m is the spherical harmonic coefficient.

Spherical Harmonics and SH Coefficients

Spherical harmonics are basis functions defined on the sphere:

$$Y_l^m(\theta, \phi)$$

$$Y_l^m(\theta,\varphi) = \begin{cases} \sqrt{2}\, K_l^m \cos(m\varphi)\, P_l^m(\cos\theta), & m > 0 \\ \sqrt{2}\, K_l^m \sin(-m\varphi)\, P_l^{-m}(\cos\theta), & m < 0 \\ K_l^0 P_l^0(\cos\theta), & m = 0 \end{cases}$$

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} \left[(x^2 - 1)^n\right]$$

$$P_l^m(x) = (-1)^m (1 - x^2)^{m/2} \frac{d^m}{dx^m} (P_l(x))$$

$$K_l^m = \sqrt{\frac{(2l+1)}{4\pi} \frac{(l-|m|)!}{(l+|m|)!}}$$

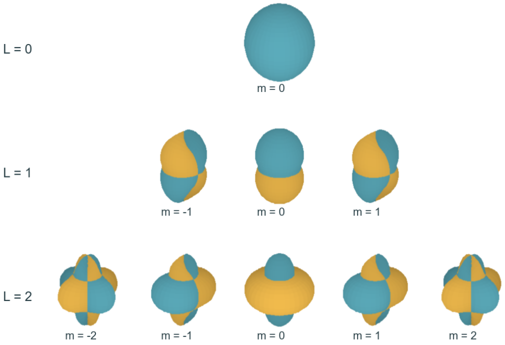

Figure: SH basis functions arranged by degree L

Spherical Harmonics and SH Coefficients

The coefficient measures how much each basis function contributes:

$$c_l^m = \int_0^{2\pi} \int_0^{\pi} f(\theta, \phi) \overline{Y_l^m(\theta, \phi)} \sin\theta \, d\theta \, d\phi$$

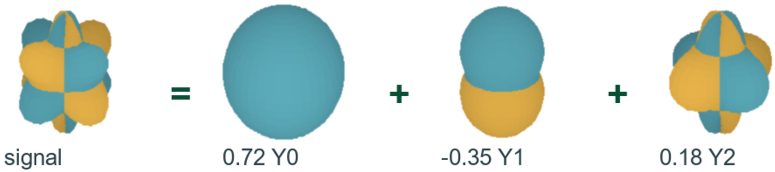

Figure: A signal reconstructed as weighted SH basis functions

Increasing SH Degree Improves Approximation

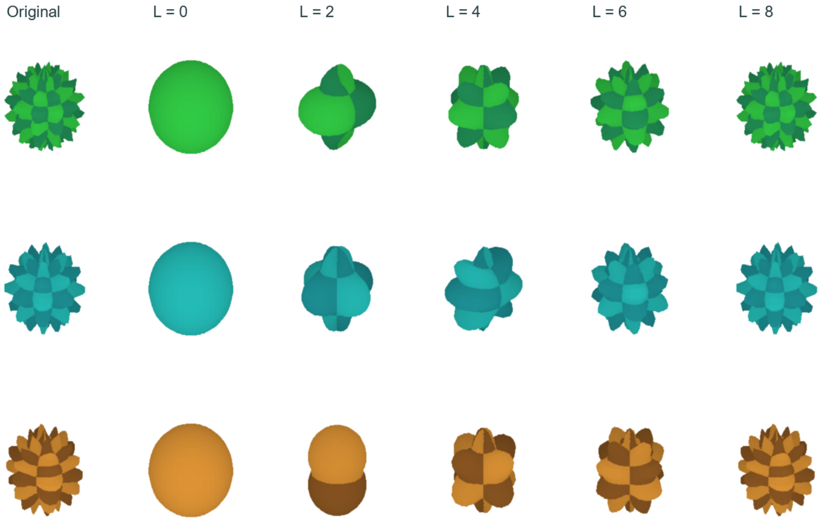

Figure: Higher SH degree captures finer angular detail

Section III: Directional Color as a Visual Signal

From color values to signal representation

Color Can Be Represented as a Signal

- Color is not only a visual property. It can also be treated as a measurable signal.
- For a fixed image point, the color value can be written as:

$$C = (R, G, B)$$

- If color changes with position, it becomes a spatial signal:

$$C(x, y) = \big(R(x, y), G(x, y), B(x, y)\big)$$

- If color changes with viewing direction, it becomes a directional signal:

$$C(\theta, \phi)$$

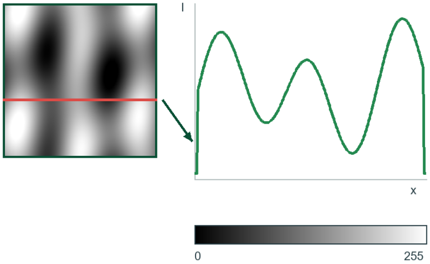

Figure: Gray values represented as a signal

From Directional Distance to Color Signal

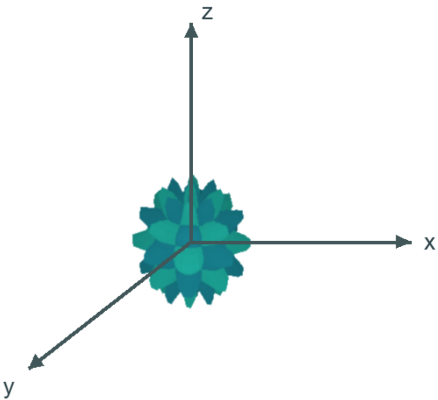

Figure: Directional distance signal

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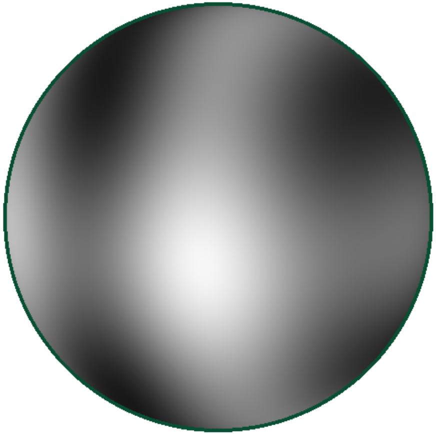

Figure: Color-mapped signal on the sphere

Multi-Channel Signals and Directional Anisotropy

- Color can be represented by three directional channels:

$$C(\theta,\phi) = \big(R(\theta,\phi), G(\theta,\phi), B(\theta,\phi)\big)$$

- Brightness, transparency, and other visual attributes can also be modeled as signals:

$$I(\theta,\phi), \quad \alpha(\theta,\phi)$$

- If a property changes with direction, it is anisotropic.

Figure: RGB, brightness, and opacity as directional signals

Section IV: Application in 3D Gaussian Splatting

Using spherical harmonics to model view-dependent appearance

SH Color Representation

Spherical harmonics are used to represent directional color:

$$\mathbf{C}_i(\boldsymbol{d}) = \sum_{l=0}^{L} \sum_{m=-l}^{l} \mathbf{c}_{i,l}^{m} Y_l^m(\boldsymbol{d})$$

where:

- $\boldsymbol{d}$ is the viewing direction from the Gaussian to the camera.
- $Y_l^m(\boldsymbol{d})$ is the SH basis value.
- $\mathbf{c}_{i,l}^{m}$ is the learned RGB coefficient.

For RGB color:

$$\mathbf{c}_{i,l}^{m} = \left(c_{R,i,l}^{m}, c_{G,i,l}^{m}, c_{B,i,l}^{m}\right)$$

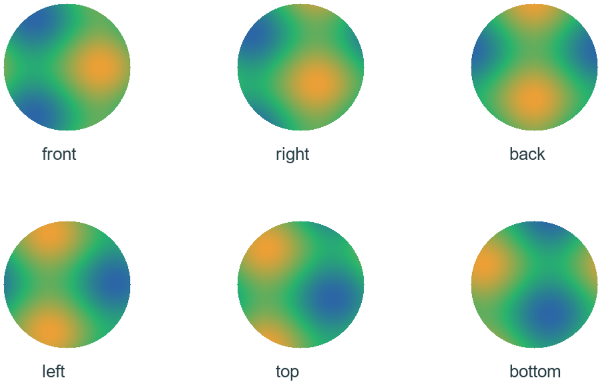

Figure: View-dependent color under SH coefficients

3D Gaussian Primitive

$$\mathcal{G}_i = \{\boldsymbol{\mu}_i, \Sigma_i, \alpha_i, \mathbf{c}_i\}$$

where:

- $\boldsymbol{\mu}_i$ is the 3D position.
- Σ_i controls anisotropic shape.
- α_i is opacity.
- $\mathbf{c}_i$ represents color information.

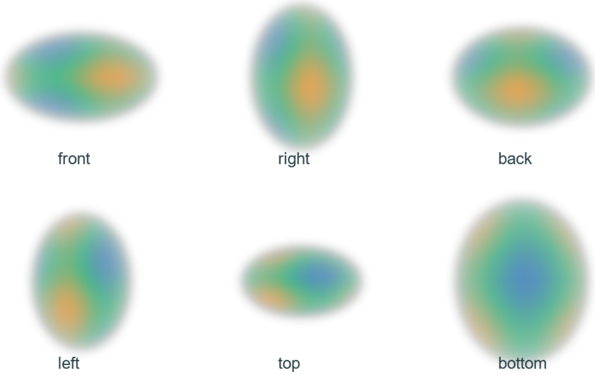

Figure: An anisotropic 3D Gaussian primitive

From One Gaussian to a 3D Scene

A single Gaussian only represents a small local primitive. By combining thousands or even millions of Gaussians, a complex object or scene can be approximated:

$$\mathcal{S} = \{\mathcal{G}_1, \mathcal{G}_2, \ldots, \mathcal{G}_N\}$$

where each Gaussian has its own:

$$\mathcal{G}_i = \{\boldsymbol{\mu}_i, \Sigma_i, \alpha_i, \mathbf{c}_i\}$$

During rendering, these Gaussians are projected to the image plane and blended together:

$$\mathbf{I}(u, v) \approx \sum_{i=1}^{N} T_i(u, v)\, \alpha_i'(u, v)\, \mathbf{C}_i(\boldsymbol{d})$$

where T_i represents accumulated transmittance from previously blended Gaussians.

The Pipeline of 3D Gaussian Splatting

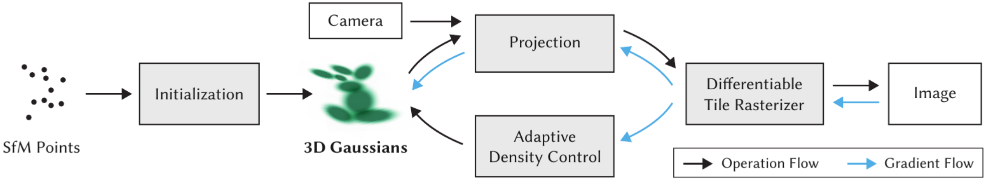

Fig. 2. Optimization starts with the sparse SfM point cloud and creates a set of 3D Gaussians. We then optimize and adaptively control the density of this set of Gaussians. During optimization we use our fast tile-based renderer, allowing competitive training times compared to SOTA fast radiance field methods. Once trained, our renderer allows real-time navigation for a wide variety of scenes.

Figure: Original optimization pipeline of 3D Gaussian Splatting

Examples of 3DGS

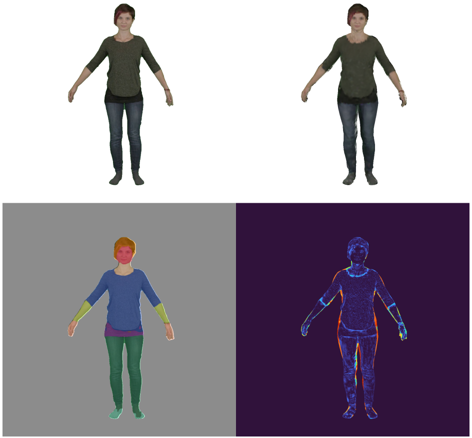

Figure: Front novel view with directional color

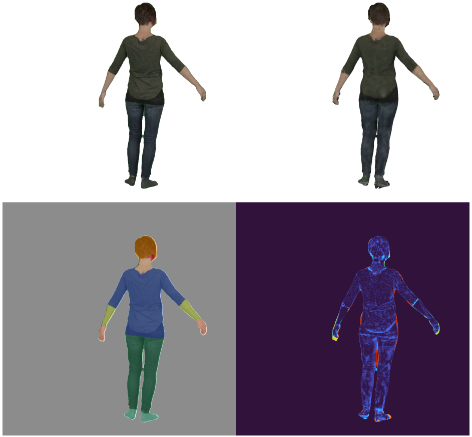

Figure: Back novel view with directional color

Section V: Demo and Evaluation

A panorama-based demonstration of spherical harmonic reconstruction

Demo Design

- The user selects a 360° panorama image from a Windows desktop GUI.
- The application computes spherical harmonic coefficients for the RGB channels.
- This demo is a simplified visualization of the directional-color idea used in 3DGS.
- The demo displays the original panorama and reconstructed results under different SH orders.

Select 360° Image

$\Downarrow$

Compute SH Coefficients

$\Downarrow$

Compare Reconstructions

Conclusion

- Fourier series show how complex signals can be represented by weighted basis functions.
- Spherical harmonics extend this idea from one-dimensional or planar signals to directional signals on a sphere.
- In 3D Gaussian Splatting, SH coefficients are used to model view-dependent RGB color.
- The panorama demo illustrates how increasing the SH order improves directional signal reconstruction.